

## ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

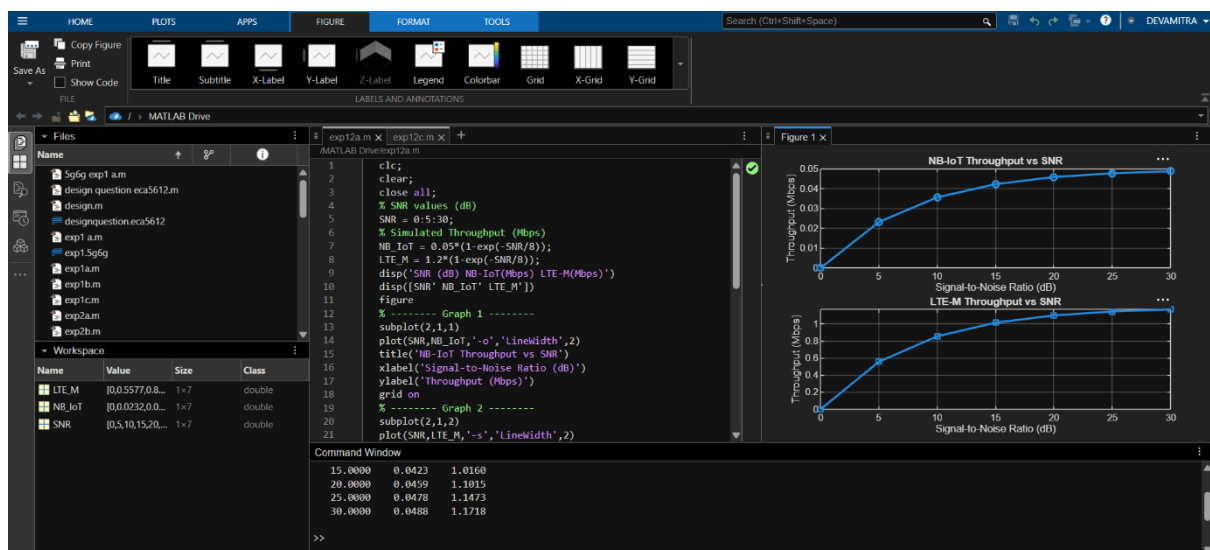
### Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

#### Objective 1 Matlab Code and Output:

##### Program:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));
disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```

## Output:



## Objective 2 Matlab Code and Output:

### Program:

```
clc;

clear;

close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])

figure

% ----- Graph 1 -----
subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
```

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(SNR,LTE\_M,'-s','LineWidth',2)

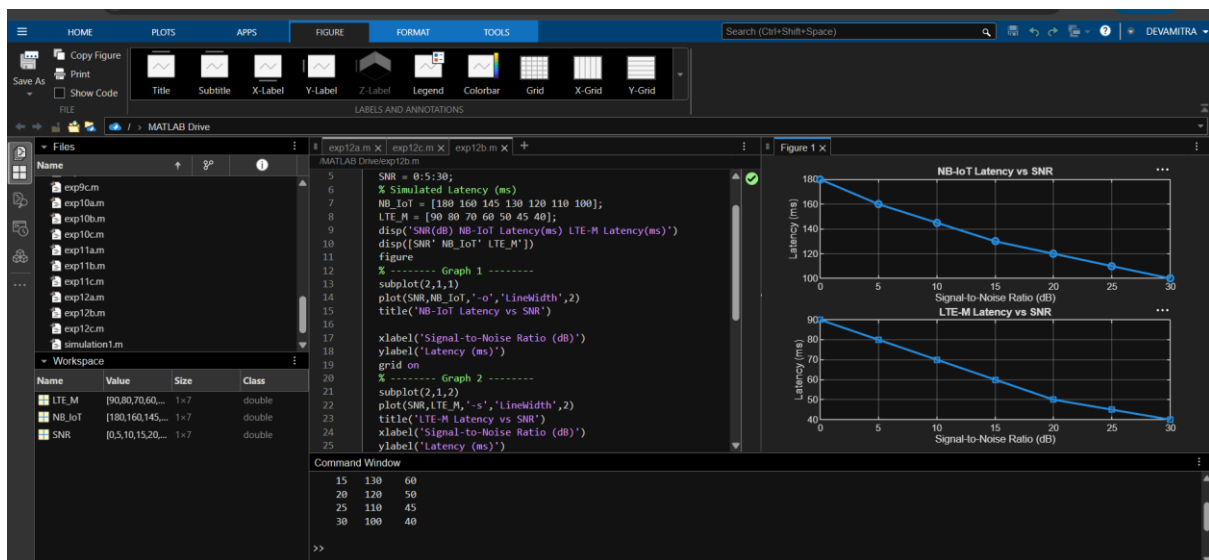
title('LTE-M Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on

**Output:**



**Objective3: Code and Output:**

**Program:**

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB\_IoT = [82 86 89 92 95 97 99];

LTE\_M = [88 91 94 96 98 99 100];

```
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%))')
```

```
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR,NB_IoT,'-o','LineWidth',2)
```

```
title('NB-IoT Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

```
title('LTE-M Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

## Output:

